

BAIJAYANTA BHATTACHARYYA

Krishnagar ◇ West Bengal
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EDUCATION

Indian Institute of Technology Khargpur

Doctor of Philosophy (Ph.D)
Department of Physics

June 2025 - Present
Coursework CGPA: 9.3/10

Indian Institute of Technology Khargpur

Master of Science
Department of Physics

July 2023 - May 2025
Overall CGPA: 9.32/10

Kalyani University

Bachelor of Science
Department of Physics Krishnagar Govt. College

Aug 2020 - July 2023
Overall CGPA : 9.8 /10

TECHNICAL STRENGTHS

Computer Languages Software & Tools

C, C++, Python
L^AT_EX, Bash Scripting, Excel, Word, Origin, Linux

EXPERIENCE

IIT Kharagpur

Doctoral Research

June 2025 - Present

- ◇ Simulate the visibilities to be observed in drift scan observations for a given telescope .
- ◇ Analyzing drift scan data of MWA to get improved upper limits of power spectrum using different techniques.
- ◇ Teaching Assistant of Course Classical Mechanics (PH31207), Autumn 2025 IIT Kharagpur. (July 2025 - Dec 2025)

LIST OF PUBLICATIONS

- ◇ The Tracking Tapered Gridded Estimator for the 21-cm power spectrum from the Murchison Widefield Array (MWA) drift scan observations – III. Improved upper limits at $z = 8.2$ from multiple pointings. Published in **Monthly Notices of the Royal Astronomical Society** <https://doi.org/10.1093/mnras/stag801>.

ACADEMIC ACHIEVEMENTS

- ◇ Cleared **CSIR-NET Physics** June-2024 Exam, Eligible for Junior Research Fellowship, **Rank-128**
- ◇ Cleared **CSIR-NET Physics** Dec-2023 Exam, Eligible for Assistant Lectureship, **Rank-52**
- ◇ Cleared **JEST 2023** Exam, **Rank-88**
- ◇ Cleared **IIT-JAM Physics Exam**, **AIR-154**
- ◇ Secured **6th** position in the **Higher Secondary Board** Exam in the State.

RELEVANT COURSES

Core Courses

Electrodynamics
Mathematical Physics
Condensed Matter Physics
Computational Physics
Classical Mechanics
Quantum Mechanics
Statistical Mechanics
Atomic And Molecular Physics
Nuclear Physics
Quantum Information and Quantum Computing
Astronomy and Astrophysics
Modern Optics
General Theory of Relativity
Group Theory And Its Applications
Physical Cosmology
Particle Physics
Quantum Field Theory
Advanced Computational Physics
Advanced Topics in Gravitation & Field Theory

Other Courses

Computational Methods Earth Science System
Physics Of Climate System
Essential Tools for Scientific Computing

EXTRA-CURRICULAR ACTIVITIES

- ◇ Represented Dept. of Physics in RSPL Football Competition, IIT KGP, May 2026. Won Gold Medal.

POSITION OF RESPONSIBILITY

Autumn 2025 Physics Courses

IIT Kharagpur

July 2025 - Dec 2025

- ◇ Teaching Assistant of Course Classical Mechanics (PH31207), Autumn 2025 .

Autumn 2026 Physics Courses

IIT Kharagpur

July 2026 - Dec 2026

- ◇ Teaching Assistant of Course TBD, Autumn 2026 .